BIOMEDICAL LIMITED ILORIN	STANDARD TESTING PROCEDURE **TITLE:** STANDARD CHEMICAL TESTING PROCEDURE FOR PARATON SYRUP (100ml)		
DEPARTMENT:	**QUALITY**	**SECTION:**	**QUALITY CONTROL**
STP NO.:	**BML/QA/STP/FPS/011/F01**	**PAGE NO.:**	**Page 1 of 8**
STP Ref.:	**BML-QDG-QA-105**	**REF. BOOK**	**BP 2024/IN-HOUSE**
EFFECTIVE DATE:	**04/01/2024**	**REVISION NO:**	**01**
REVIEW DATE:	**04/01/2026**	**SUPERSEDE:**	**00**

S/N	TEST	DESCRIPTIONS OF PROCEDURE
1.	**pH**	1. Take sample sufficient to immerse the electrode-head, and dip the electrode in the sample. 2. Wait until the reading stabilises, then take readings.
2.	**Chlorpheniramine Maleate** **Label Claim:** **2mg/5ml**	**METHOD 1:** ***Standard Solution:*** 1mg/100mL Weigh accurately 100mg of Chlorphenamine maleate reference standard. Dissolve and make up to 100mL with water. From the resulting solution, dilute 1mL to 100mL with the same solvent. ***Sample Solution:*** Dissolve 5mL of the syrup in water and make it up to 100mL. From the resulting, dilute 50mL to 100mL. **Procedure:** To 10mL of each of the each solution, add 1mL of 1% Bromothymol solution, then extract with 25mL of chloroform. Measure the absorbance of the solutions at 420nm.

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		% Assay of Chlorpheniramine Maleate = $\frac{absorbance of sample}{absorbance of standard}$x 100 **METHOD 2:** ***Standard Solution:*** 25mcg/mL of substance in 0.5N sulphuric acid. ***Sample solution:*** Take a quantity of the sample equivalent to 20mg of the compound, acidify with hydrochloric acid and extract with three 50mL portions of ether. Discard ether layer, make acidic layer alkaline with 5N NaOH solution and extract with three 50mL portions of ether. Wash combined ether extract with water till free of alkali. Extract combined ether layer with three 50mL portions of 0.5N sulphuric acid. Dilute the acidic extract to 200mL (100mcg/mL). Dilute further appropriately with 0.5N sulphuric acid to get 20mcg/mL. Measure the absorbance of standard and sample solutions at maximum at about 264nm using 0.5N sulphuric acid as blank and calculate the contents of Chlorpheniramine Maleate by comparison. % Assay of Chlorpheniramine Maleate = $\frac{absorbance of sample}{absorbance of standard}$x 100
3.	**Paracetamol** **Label Claim:**	**METHOD 1**: Paracetamol ***Standard Solution:*** 1mg/100mL

	120mg/5ml	• Weigh accurately and dissolve 100mg of standard paracetamol in 0.1M NaOH, and make up the volume to 100mL • From the resulting solution, take 1mL to 100mL with the same solvent ***Sample Solution:*** • Dissolve an equivalent of 100mg of the syrup (4mL) of the syrup in 0.1mL and make up the volume to 100mL. • From the resulting solution, take 1mL to 100mL with the same solvent • Read absorbance at 257nm and compare results with standard % Assay of Paracetamol = $\frac{absorbance of sample}{absorbance of standard}$x 100 **NOTE:** in acidic solution, paracetamol has lambda max 246nm, whereas in alkaline solution, it has lambda max 257nm. This bathochromic shift 246→257 nm is due to molecular re-arrangement in the solvent. ***REFERENCE:*** Sethi's Quantitative Analysis of Pharmaceutical Formulations. Vol. 2
4.	**Ascorbic acid** **Label Claim:**	**METHOD 1** **Procedure:** Weigh accurately, syrup equivalent to 40mg of ascorbic acid. Add to this, 10mL acetic acid and 20mL water. Titrate with 0.1N iodine solution using starch as indicator. Each mL of 0.1N iodine solution is equivalent to 0.008806g of ascorbic

	40mg/5ml ***Note: 55mg due to overage***	acid. $\% \boldsymbol{Assay} = \frac{(Average\ Titre)\ x\ Standard\ Factor\ x\ 8.81}{55}\ x\ 100$ **Note:** *Iodine is a mild oxidizing agent. It oxidizes ascorbic acid to dehydroascorbic acid and endpoint being determined by the formation of blue-coloured iodine-starch inclusion complex.* **METHOD 2:** ***Standard*** Weigh accurately 150mg of Ascorbic Acid and dissolve in 0.01M HCl and make up to mark using a 100ml volumetric flask. Take exactly 1ml of the resulting solution and make up to mark of a 100ml volumetric flask using the same solvent. ***Sample*** Take quantitatively equivalent of 150mg (exactly 5ml) of the Syrup and dissolve with 0.01M HCl and make up to mark using a 100ml volumetric flask. Take accurately 1ml of the resulting solution and make up to mark of a 100ml volumetric flask using the same solvent. % Assay ofAscorbic Acid = $\frac{absorbance of sample}{absorbance of standard}$x 100 ***REFERENCE:*** *Sethi's* Quantitative Analysis of Pharmaceutical Formulations. Vol. 3

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		METHOD 3: **Procedure:** Weigh accurately, syrup equivalent to 40mg of ascorbic acid, add 10mL of dilute sulphuric acid and then 80mL of carbon dioxide-free water. Add 1mL of starch indicator. Titrate with 0.05M iodine until a persistent violet-blue colour is obtained. 1mL of 0.05M iodine is equivalent to 8.81mg of $C_6H_8O_6$. $$\%\ \boldsymbol{Assay} = \frac{(Average\ Titre)\ x\ Standard\ Factor\ x\ 8.81}{55}\ x\ 100$$ **REFERENCE:** *British Pharmacopoeia, 2024, Ascorbic acid: Ph. Eur. Monograph 0253.*
5.	**Microbial Enumeration Test**	- Determination of total aerobic microbial count (TAMC) and total combined yeasts and molds count (TYMC) by the pour plate method. - Dispense 1 ml of diluted sample in the center of the Petri plate using a sterile micropipette or calibrated pipette. - Open the lid of the prepared media, flame its mouth. - Pour about 15 mL of sterilized molten media at the appropriate temperature above the sample. - Close the lid of the plate then mix the sample in the media properly by gently

swirling the plate. The plate is generally swirled in an "S" or "8" shape.

- Allow the media to completely solidify.
- Incubate the plate in an inverted position under suitable incubation conditions (Incubate the plates of Soybean Casein Digest Agar at 30°C-35°C for 3-5days for TAMC and the plates of Sabouraud Dextrose Agar at 20°C to 25°C for 5-7 days.

TEST FOR ABSENCE OF PATHOGEN

- Measure 10ml /10g of the samples, dilute 1n 90ml of nutrient broth
- Transfer 10ml into 90ml of Tryptone soya broth
- Incubate at 30-35^{0}c for 18-24hours
- Transfer 1 ml of the dilution into 100ml of MacConkey broth and incubate at 42-44°c in 18-72hours
- Subculture on MacConkey Agar or Eosin Methylene blue Agar and incubate at 30-35°c
- For negative control, proceeding same as above except without sample, streak a portion of incubated sterile MacConkey broth medium on the surface of Eosin Methylene Blue Agar.
- Incubate plates at 30 to 35° for 48 hours.

A. Result Interpretation

- Following the incubation, developed colonies are observed. Each colony will account for one viable microbial cell or one colony-forming unit.
- If all the colonies are of the same type, then we can predict that the sample contained only one type of microbial genera.
- However, if there are different genera or species producing similar types of colonies, it is required to carry out identification test.
- If the colonies are of different morphology, then we can conclude that the sample contained a mixed population.

Count the colonies and calculate CFU/mL by using the formula:

$$\text{CFU/mL} = \frac{\text{Total number of colonies obtained x dilution factor}}{\text{Volume of sample used (aliquot)}}$$

- This will give the total number of viable microbial cells present in the given sample.
- For optimum count, the number of colonies must be between 20 – 200 CFU/mL. Beyond this limit, the whole procedure must be repeated.
- If the number of the colony is less than 20, it is suggested to use the sample of lower dilution, whereas, if the total number of colonies exceeds 200, it is suggested to use the sample of higher dilution on successive repeats.

		- The result should be recorded following the Standard testing procedure SOP **BML-QDG-QA-105** - If the colonies are fused or the whole plate is covered with a single colony, then report as "too numerous to count" (TNTC) and repeat the process of taking the sample at a higher dilution.

CURRENT REVISION	SUPERSEDE	REASON FOR REVIEW
00	NIL	FIRST ISSUE
01	00	Review of the procedures Change in header and footer format

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